DTC P0016:00 [SKYACTIV-D 1.5]

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DTC P0016:00	Camshaft position/Crankshaft position correlation problem
DETECTION CONDITION	• CKP sensor and CMP sensor input signal does not match while the crankshaft rotates **15 times** when the following condition A or B is met: **Condition A:** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine on) — Engine speed: **1,000 rpm or less** — The following DTCs are not detected: • CKP sensor: P0336:00, P0337:00 • CMP sensor: P0341:00, P0342:00 **Condition B:** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine on) — Engine speed: **700—1,000 rpm** — Engine coolant temperature: **−10 °C {14 °F} or more** (engine on) — The following DTCs are not detected: • ECT sensor: P0117:00, P0118:00 • CKP sensor: P0336:00, P0337:00 • CMP sensor: P0341:00, P0342:00 **Diagnostic support note** • This is a continuous monitor (CCM). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• PCM restricts engine-transaxle integration control. (ATX) • Inhibits engine-stop by operating the i-stop function.
POSSIBLE CAUSE	• CMP sensor connector or terminals malfunction • CKP sensor connector or terminals malfunction • PCM connector or terminals malfunction • Deviation between camshaft and CMP sensor detection area • Damage to detection area of CMP sensor • CMP sensor malfunction • CKP sensor pulse wheel malfunction • CKP sensor malfunction • Improper valve timing — Improper installation of timing chain to the camshaft sprocket teeth — Timing chain damage — Camshaft sprocket damage • PCM malfunction
SYSTEM WIRING DIAGRAM	Not applicable

Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.

STEP	INSPECTION		ACTION
3	**VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any other PENDING CODEs and/or DTCs present?	Yes	Go to the applicable PENDING CODE or DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
4	**INSPECT CMP SENSOR CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the CMP sensor connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 10.
		No	Go to the next step.
5	**INSPECT CKP SENSOR CONNECTOR CONDITION** • Disconnect the CKP sensor connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 10.
		No	Go to the next step.
6	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 10.
		No	Go to the next step.
7	**INSPECT CMP SENSOR** • Inspect the CMP sensor. (See CAMSHAFT POSITION (CMP) SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the CMP sensor, then go to Step 10. (See CAMSHAFT POSITION (CMP) SENSOR REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
8	**INSPECT CKP SENSOR** • Inspect the CKP sensor. (See CRANKSHAFT POSITION (CKP) SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the CKP sensor, then go to Step 10. (See CRANKSHAFT POSITION (CKP) SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
9	**INSPECT TIMING CHAIN INSTALLATION CONDITION** • Verify the condition of the timing chain assembly (valve timing, looseness, jumping). (See TIMING CHAIN REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results. Assemble the timing chain using the correct timing, then go to the next step. (See TIMING CHAIN REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
10	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Perform the KOER self test. (See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
11	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.